

products will reduce reliance on local demand.

"Most companies have quality standards such as ISO and LCSO in place but these certifications do not have much impact on the actual product quality. I think it is high time for Indian companies to focus on Japanese quality standards and follow those like cult," opines Muniswamy.

On the same lines, Doshi mentions that IPC standards are and will be the core standards to follow. He says, "More and more customers are now demanding higher level of quality standards, parameters and systems prevalent in the automotive (TS), aerospace (AS) and medical (ISO 13485) arena."

Growing inorganically

Almost all senior thought leaders who shared their insights for this survey believe in the possibility of inorganic growth in this segment, which could either be in the form of mergers and acquisitions (M&As) or through joint ventures (JVs). For example, Muniswamy states, "I strongly believe in M&As as these help companies to grow inorganically in a short span of time. Entrepreneurs often fail to grow beyond a point due to lack of vision and leadership. They develop strong resistance to change. Most early starters restrict themselves to the domestic market, thereby limiting their exposure, unlike Chinese firms that conquer world markets. There are still many PCB companies that are running businesses like a lifestyle and these are good cases for M&As. These companies need to let capable people run the company and original founders should sit back and enjoy the success."

However, Doshi opines, "Globally, most PCB industries that have survived the Chinese onslaught and those in China also are mostly owner-managed. The need for a huge skill set (technological, managerial and financial) required in a professional who can successfully grow the company is largely absent in India." He

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Favourable conditions

- Special incentive schemes such as MSIPs from DeitY
- Electronics manufacturing clusters being set up by associations such as ELCINA
- Special PCB manufacturing cluster by Madhya Pradesh government in Gwalior
- Special schemes announced by Orissa government for the manufacturing sector
- Special electronics manufacturing clusters being set up in Andhra Pradesh

adds, "Given the relatively small sizes of most companies in the PCB industry, although theoretically logical, a change may end up costing the current opportunity of growth on hand."

Since most PCB manufacturers in the country are incapable of investments required for producing high-tech products for devices like set-top boxes and mobile handsets in required volume, any kind of partnership with foreign companies could help Indian companies manufacture high-tech PCBs by leveraging technology transfer.

India advantages

In the last few months, campaigns like Make in India and Digital India have attracted huge FDI inflows to the country from across the globe. This has created more opportunities for outsourcing work by companies such as Foxconn. Exponential growth of the mobile phone market is fueling the growth of electronics manufacturing units, and the day is not far when a large company would set up a manufacturing facility for HDI boards for the mobile phone industry.

"For the first time in decades, venture capital (VC) funds are available with as low as six per cent to eight per cent interest rates. Some companies are looking for JVs with foreign companies to bring cutting-

edge technology to India. There are at least half a dozen companies in the pipeline for establishing green-field PCB projects in India, which include two JVs. Therefore right now is the best time for investment in the PCB industry in India," says Muniswamy.

Looking forward

Despite limitations like capital intensive, low investment-to-sales ratio, logistics and infrastructural issues, the Indian PCB industry should be able to witness sustainable growth if the government can provide a level playing field for the industry. As Vasani rightly comments, "We do not need nurturing or subsidy. Instead, the government should be a catalyst for growth. Take the case of the Indian automotive industry for example; the government discourages automotive import with high duty structures and complicated registration process, compelling automotive companies to manufacture in India, resulting in a proper manufacturing ecosystem. The government should create a similar environment for PCBs and other electronic components as well."

Given that there is huge domestic demand for electronic products, it will help not just the PCB industry but the entire electronic ecosystem to achieve long-term sustainable growth in-line with Make in India. ●